MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)

FIFTH SEMESTER EXAMINATION

(2024 Admission Onwards)

MG5DSEMAT301: MATHEMATICAL MUSINGS BEYOND CLASSROOM

Duration: 2 Hours **Maximum Marks: 70**

(Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability)

Part A

Answer any five questions. Each question carries 2 marks.

1. How did the Indian Mathematician Brahmagupta define zero as a number and placed in the number line? [U][2]
2. What mathematical challenge did Paul Erdos pose to the little boy Louis Posa over lunch and how did Posa solve it? [K][5]
3. What explanation did Ramanujan give to G.H. Hardy about his mathematical formulas? [U][3]
4. Mention the three African American women mathematicians whose real life contributions at NASA are depicted in the film "Hidden Figures". [K][4]
5. Mention any two major texts (and the respective authors) that constitute the primary evidence of the work of Kerala School of Mathematics. [K][2]
6. Write the Madhava-Gregory Series for the inverse tangent and deduce the Liebnitz series for $\frac{\pi}{4}$. [K][1]
7. How does the humanist philosophy distinguish between "Counting Numbers" and "Pure Numbers"? [An] [7]
8. State Cantor's theorem. [K][1]

Part B

Answer any five questions. Each question carries 2 marks.

9. Describe the mathematical consequences of performing the basic operations (addition, multiplication and division) with zero. [U][2]
10. Explain the following principle using Galton board: "things that seem hopelessly random and unpredictable when viewed in isolation often turn out to be lawful and predictable when viewed in aggregate". [A][3]

11. Discuss the ethical dilemma faced by Alan Turing's team immediately after breaking the Enigma code and explain how they resolved it. [E][5]
12. Write a short note on an introduction to mathematics on the silver screen. [An] [4]
13. Explain the Madhava's rule for generating the power series for Indian Sine and Indian Versine. Also write the series. [U][2]
14. Analyze the navigational and calendrical motivations that drove European interest in Indian scientific knowledge during the 16th and 17th centuries. [An] [6]
15. Discuss how the mathematical understanding of the number "2" has changed and evolved over time. [An] [5]
16. Compare formalism, logicism and intuitionism. [An] [7]

Part C

Answer any three questions. Each question carries 10 marks.

17. Explain Page Rank and the Page Rank Algorithm using examples. [A][6]
18. Write about the mathematical legacy of John Nash presented in the film "A Beautiful Mind". [U][2]
19. Based on the film "The man who knew infinity", describe the early life of Srinivasa Ramanujan in India and discuss the challenges, cultural barriers and health struggles, he faced while living in England. [An] [4]
20. Discuss the socio-cultural background and institutional environment that enabled the development of the Kerala School of Mathematics. [An] [7]
21. "Whether Mathematics is created or discovered"-Critically evaluate the Platonists, formalists and humanist views using examples. [An] [5]
22. Explain Paul Benacerraf's under determination problem regarding the reduction of natural n-umbers to set- theoretic constructions. [An] [5]